LAB EXPERIMENTS

TOOL:R Programming **RegisterNo:192424335**

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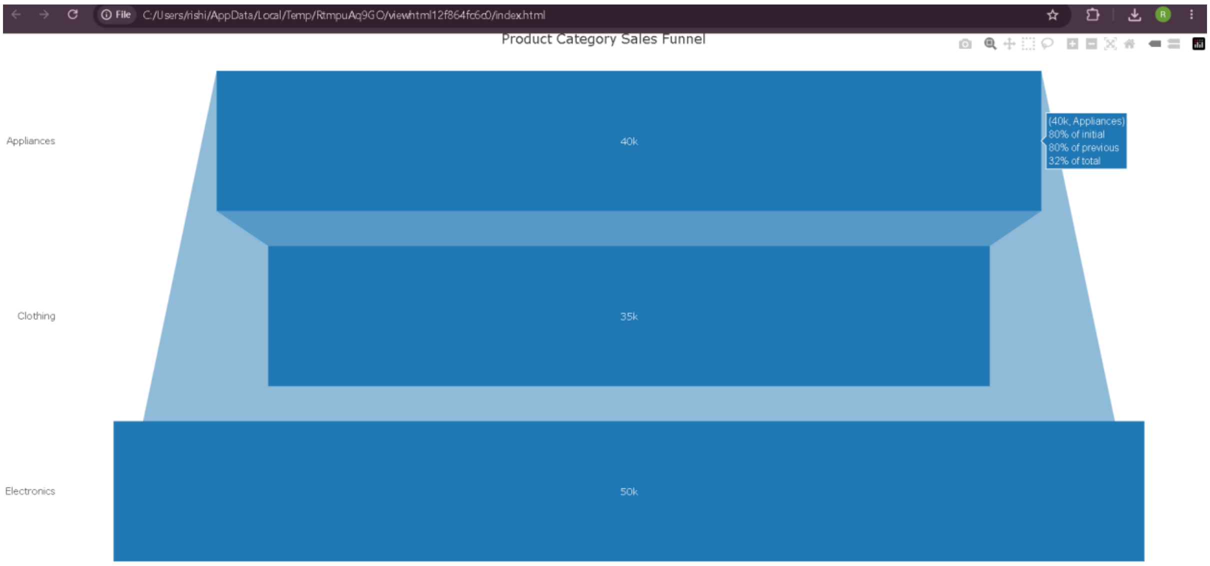
COURSE:Data Handling and Visualization **Code: DSA0604**

SET-11

Q1- Funnel Chart

R PROGRAM:

```
library(plotly)
sales <- data.frame(
  Category = c("Electronics", "Appliances", "Clothing"),
  Sales = c(50000, 40000, 35000)
)
plot_ly(
  type = "funnel",
  y = sales$Category,
  x = sales$Sales
) %>%
  layout(title = "Product Category Sales Funnel")
```



Q2 -Table

R PROGRAM:

```
sales <- data.frame(
  Category = c("Electronics", "Clothing", "Appliances"),
  Sales = c(50000, 35000, 40000)
)
print(sales)
```

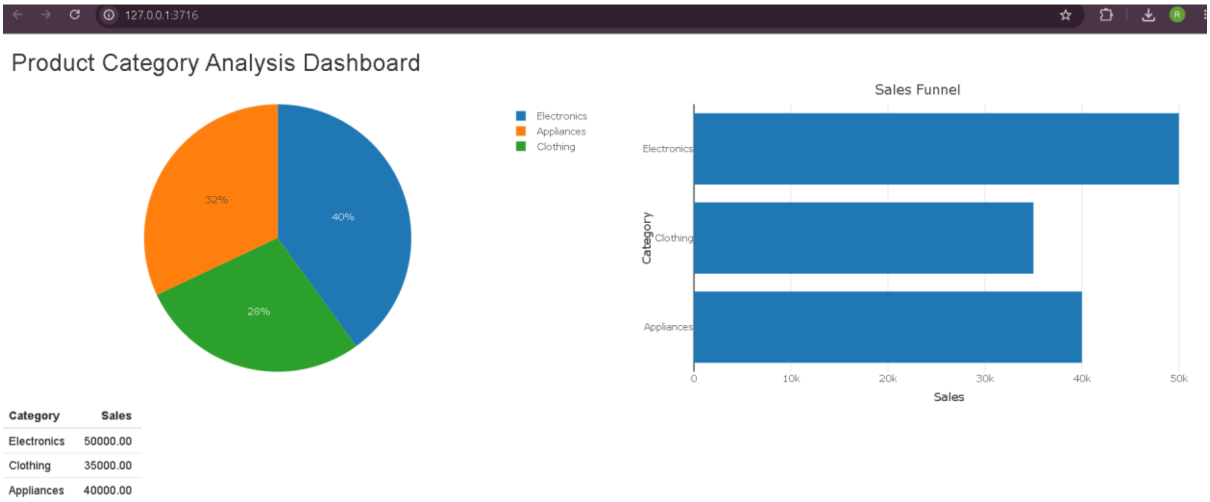
```
+ )
>
> print(sales)
     Category Sales
1 Electronics 50000
2    Clothing 35000
3  Appliances 40000
> s
```

Q3- Shiny Dashboard in R)

R PROGRAM:

```
library(shiny)
library(plotly)
sales <- data.frame(
  Category = c("Electronics", "Clothing", "Appliances"),
  Sales = c(50000, 35000, 40000)
)
ui <- fluidPage(
  titlePanel("Product Category Analysis Dashboard"),
  fluidRow(
    column(6, plotlyOutput("pie")),
    column(6, plotlyOutput("funnel"))
  ),
  fluidRow(
    tableOutput("table"))
)
```

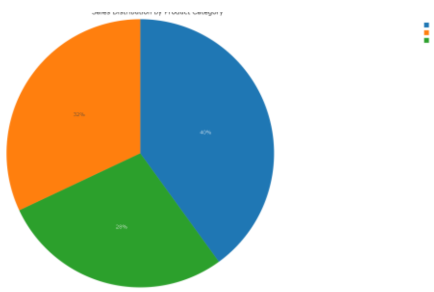
```
server <- function(input, output) {
  output$pie <- renderPlotly({
    plot_ly(
      data = sales,
      labels = ~Category,
      values = ~Sales,
      type = "pie"
    )
  })
  output$funnel <- renderPlotly({
    plot_ly(
      data = sales,
      x = ~Sales,
      y = ~Category,
      type = "bar",
      orientation = "h"
    ) %>%
      layout(title = "Sales Funnel")
  })
  output$table <- renderTable({
    sales
  })
}
shinyApp(ui, server)
```



Q4- Pie Chart

R PROGRAM:

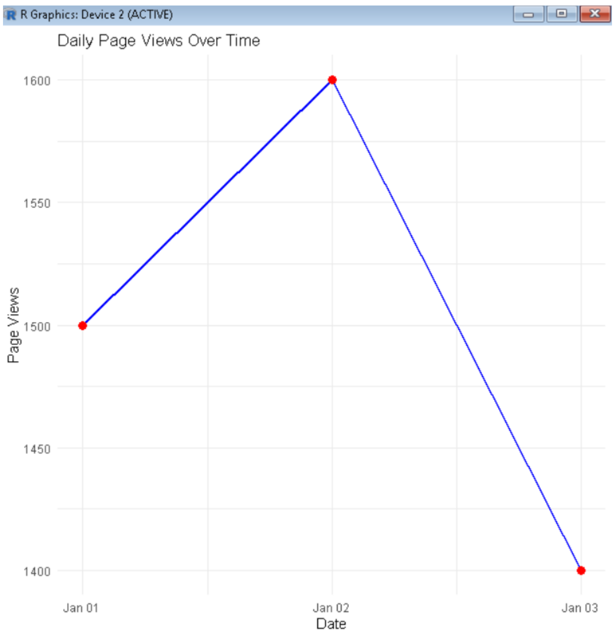
```
library(plotly)
sales <- data.frame(
  Category = c("Electronics", "Clothing", "Appliances"),
  Sales = c(50000, 35000, 40000)
)
plot_ly(
  sales,
  labels = ~Category,
  values = ~Sales,
  type = "pie"
) %>%
  layout(title = "Sales Distribution by Product Category")
```



SET-12

Question 1: Line Chart

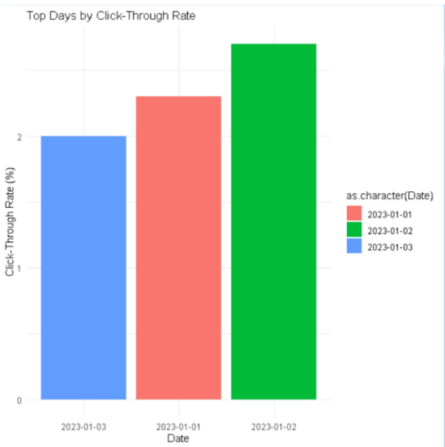
```
library(ggplot2)
traffic <- data.frame(
  Date = as.Date(c("2023-01-01","2023-01-02","2023-01-03")),
  PageViews = c(1500,1600,1400),
  CTR = c(2.3,2.7,2.0)
)
ggplot(traffic, aes(x = Date, y = PageViews, group = 1)) +
  geom_line(color = "blue", linewidth = 1) +
  geom_point(size = 3, color = "red") +
  labs(
    title = "Daily Page Views Over Time",
    x = "Date",
    y = "Page Views"
  ) +
  theme_minimal()
```



Q2- Bar Chart (Top N Click-Through Rates)

R PROGRAM:

```
library(ggplot2)
traffic <- data.frame(
  Date = as.Date(c("2023-01-01","2023-01-02","2023-01-03")),
  PageViews = c(1500,1600,1400),
  CTR = c(2.3,2.7,2.0)
)
topN <- traffic[order(-traffic$CTR), ]
ggplot(topN, aes(x = reorder(as.character(Date), CTR), y = CTR, fill = as.character(Date))) +
  geom_bar(stat = "identity") +
  labs(
    title = "Top Days by Click-Through Rate",
    x = "Date",
    y = "Click-Through Rate (%)"
  ) +
  theme_minimal()
```

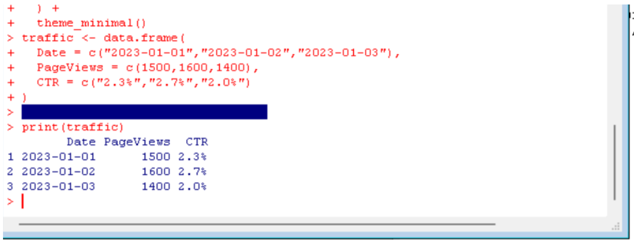


Question 3: Table

R PROGRAM:

```
traffic <- data.frame(
  Date = c("2023-01-01","2023-01-02","2023-01-03"),
  PageViews = c(1500,1600,1400),
```

```
  CTR = c("2.3%","2.7%","2.0%")
)
print(traffic)
```

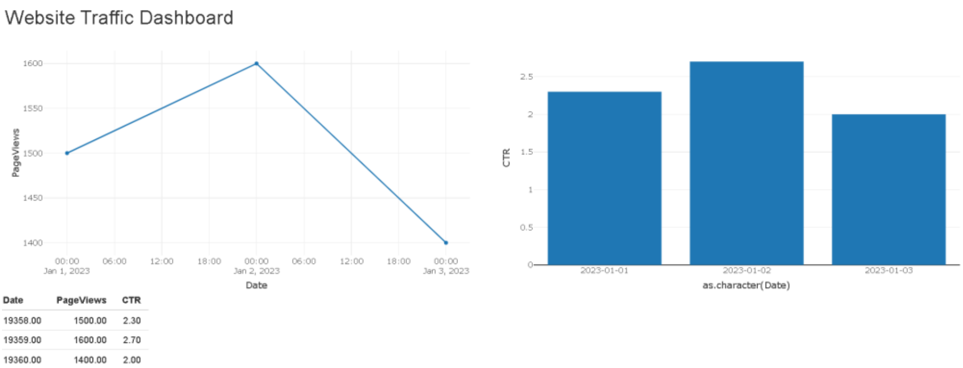


```
+   ) +
+   theme_minimal()
> traffic <- data.frame(
+   Date = c("2023-01-01","2023-01-02","2023-01-03"),
+   PageViews = c(1500,1600,1400),
+   CTR = c("2.3%","2.7%","2.0%")
+ )
> [redacted]
> print(traffic)
        Date PageViews  CTR
1 2023-01-01      1500 2.3%
2 2023-01-02      1600 2.7%
3 2023-01-03      1400 2.0%
>
```

Q4-Dashboard (Shiny)

```
library(shiny)
library(plotly)
traffic <- data.frame(
  Date = as.Date(c("2023-01-01","2023-01-02","2023-01-03")),
  PageViews = c(1500,1600,1400),
  CTR = c(2.3,2.7,2.0)
)
ui <- fluidPage(
  titlePanel("Website Traffic Dashboard"),
  fluidRow(
    column(6, plotlyOutput("lineChart")),
    column(6, plotlyOutput("barChart"))
  ),

  fluidRow(
    tableOutput("trafficTable")
  )
)
server <- function(input, output) {
  output$lineChart <- renderPlotly({
    plot_ly(
```

```
      traffic,
      x = ~Date,
      y = ~PageViews,
      type = "scatter",
      mode = "lines+markers"
    )
  })
  output$barChart <- renderPlotly({
    plot_ly(
      traffic,
      x = ~as.character(Date),
      y = ~CTR,
      type = "bar"
    )
  })
  output$trafficTable <- renderTable({
    traffic
  })
}
shinyApp(ui, server)
```

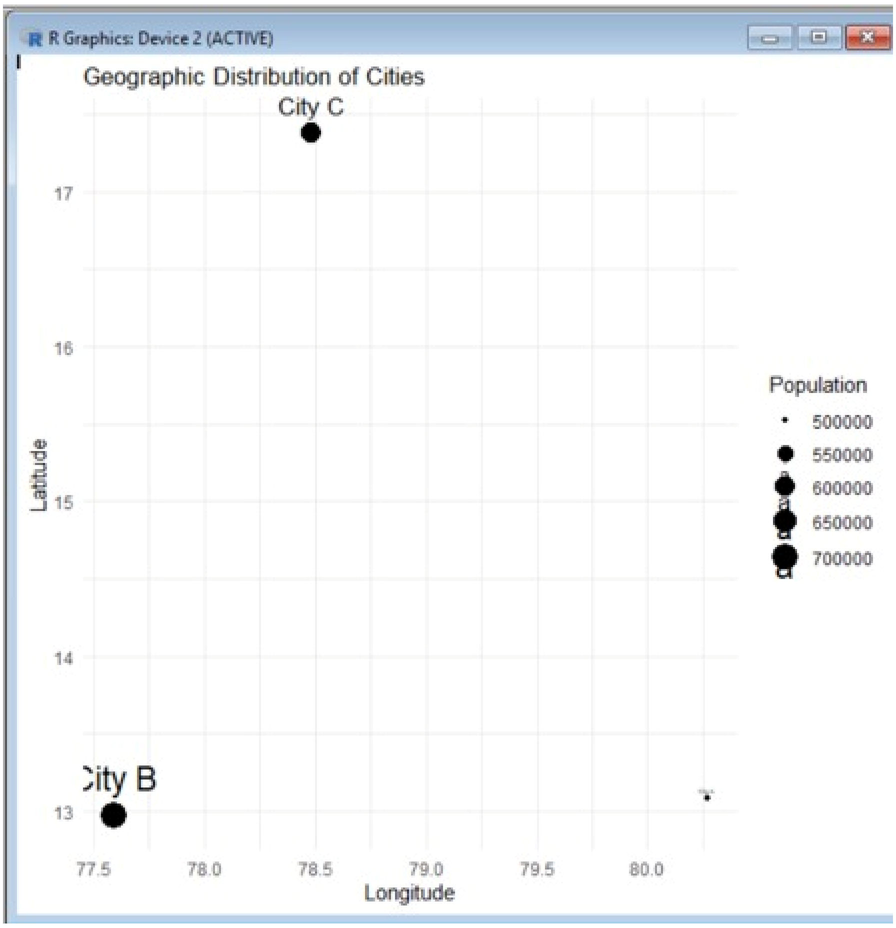


SET-13

Q1 Map Chart

R program:

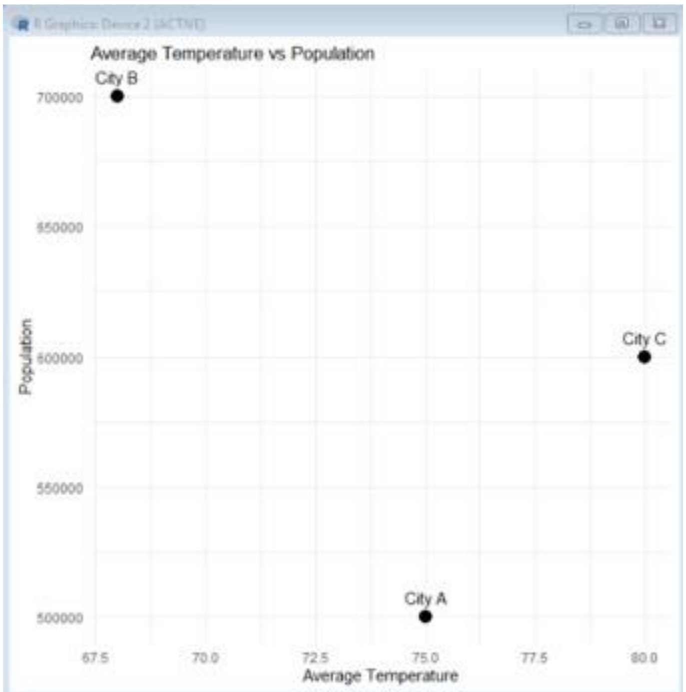
```
library(leaflet)
geo <- data.frame(
  City = c("City A","City B","City C"),
  Latitude = c(13.0827,28.7041,19.0760),
  Longitude = c(80.2707,77.1025,72.8777)
)
leaflet(data = geo) %>%
  addTiles() %>%
  addMarkers(
    lng = geo$Longitude,
    lat = geo$Latitude,
    popup = geo$City
  )
```



Q2- Scatter Plot

R PROGRAM:

```
library(ggplot2)
geo <- data.frame(
  City = c("City A","City B","City C"),
  Population = c(500000,700000,600000),
  Temperature = c(75,68,80),
  Elevation = c(1000,800,1200)
)
ggplot(geo, aes(x = Population, y = Temperature, label = City)) +
  geom_point(size = 4, color = "blue") +
  geom_text(vjust = -1) +
  labs(
    title = "Average Temperature vs Population",
    x = "Population",
    y = "Average Temperature"
  ) +
  theme_minimal()
```



Q3 TABLE

R PROGRAM:

```
geo <- data.frame(
  City = c("City A","City B","City C"),
  Population = c(500000,700000,600000),
  Temperature = c(75,68,80),
  Elevation = c(1000,800,1200)
)
print(geo)
```

```
> print(geo)
    City Population Average_Temperature Elevation
1 City A      5e+05                  75      1000
2 City B      7e+05                  68       800
3 City C      6e+05                  80      1200
> 
```

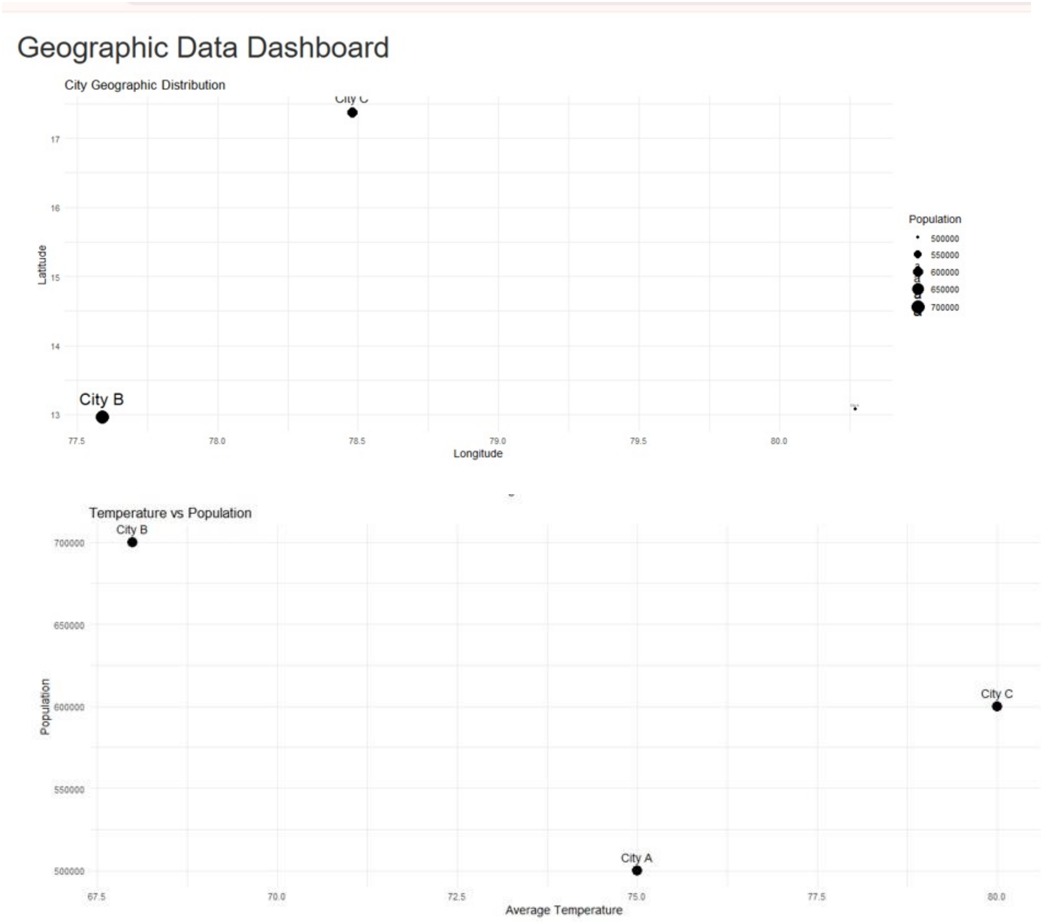
Q4- Dashboard (Shiny)

R -PROGRAM:

```
library(shiny)
library(plotly)
library(leaflet)
geo <- data.frame(
  City = c("City A","City B","City C"),
  Latitude = c(13.0827,28.7041,19.0760),
  Longitude = c(80.2707,77.1025,72.8777),
  Population = c(500000,700000,600000),
  Temperature = c(75,68,80),
  Elevation = c(1000,800,1200)
)
ui <- fluidPage(
  titlePanel("Geographic Data Dashboard"),
```

```
  fluidRow(
    column(6, leafletOutput("map")),
    column(6, plotlyOutput("scatter"))
  ),
  fluidRow(
    tableOutput("table")
  )
)
server <- function(input, output){
 output$map <- renderLeaflet({
    leaflet(geo) %>%
      addTiles() %>%
      addMarkers(
        lng = ~Longitude,
        lat = ~Latitude,
        popup = ~City
      )
  })
  output$scatter <- renderPlotly({
    plot_ly(
      geo,
      x = ~Population,
      y = ~Temperature,
      type = "scatter",
      mode = "markers+text",
      text = ~City
    )
  })
  output$table <- renderTable({
    geo
```

```
  })
}
shinyApp(ui, server)
```



City	Population	Temperature	Elevation	Longitude	Latitude
City A	500000.00	75.00	1000.00	80.27	13.08
City B	700000.00	68.00	800.00	77.59	12.97
City C	600000.00	80.00	1200.00	78.48	17.38

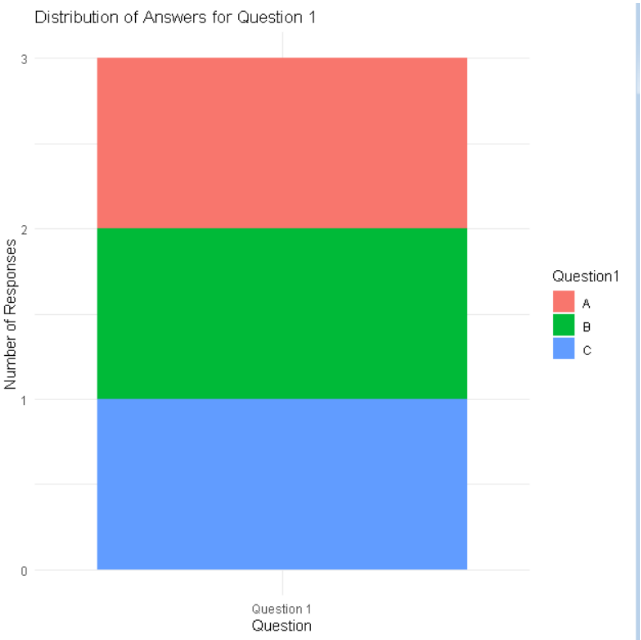
SET-14

Q1- Stacked Bar Chart

R PROGRAM:

```
library(ggplot2)
survey <- data.frame(
  Respondent = c(1,2,3),
  Question1 = c("A","B","C"),
  Question2 = c("B","A","A"),
```

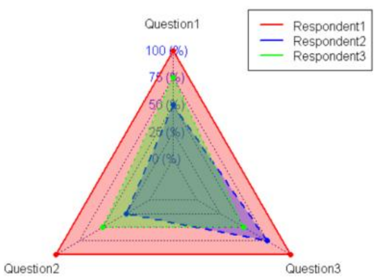
```
  Question3 = c("C","D","B")
)
ggplot(survey, aes(x = "Question 1", fill = Question1)) +
  geom_bar() +
  labs(
    title = "Distribution of Answers for Question 1",
    x = "Question",
    y = "Number of Responses"
  ) +
  theme_minimal()
```



Q2- Radar Chart

```
library(fmsb)
survey <- data.frame(
  Q1 = c(4,2,3),
  Q2 = c(4,1,2),
  Q3 = c(4,3,2)
)
radar <- rbind(
  c(4,4,4),
```

```
  c(0,0,0),
  survey
)
colnames(radar) <- c("Question1","Question2","Question3")
radarchart(
  radar,
  axistype = 1,
  pcol = c("red","blue","green"),
  pfcol = c(rgb(1,0,0,0.3), rgb(0,0,1,0.3), rgb(0,1,0,0.3)),
  plwd = 2
)
legend(
  "topright",
  legend = c("Respondent1","Respondent2","Respondent3"),
  col = c("red","blue","green"),
  lty = 1,
  lwd = 2
)
```



Q3- Table

R program:

```
survey <- data.frame(
  Respondent = c(1,2,3),
```

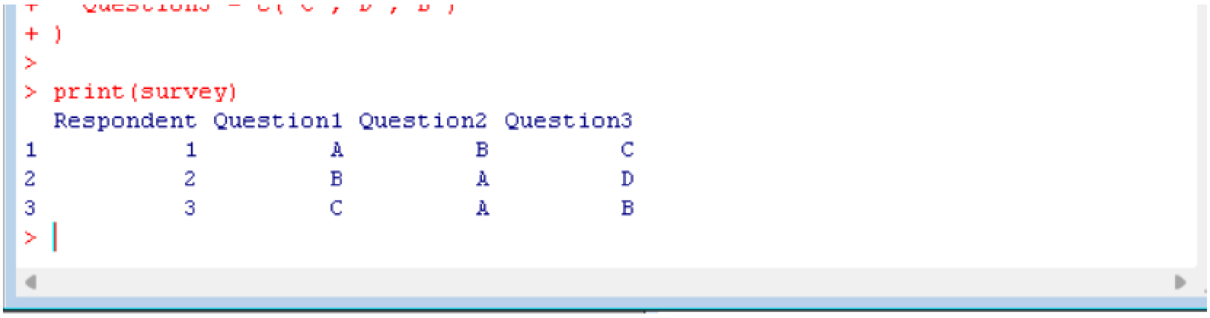
Question1 = c("A","B","C"),

Question2 = c("B","A","A"),

Question3 = c("C","D","B")

)

print(survey)

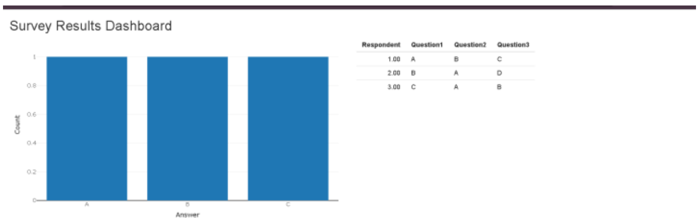


Q4-Dashboard (Shiny)

R PROGRAM:

```
library(shiny)
library(plotly)
survey <- data.frame(
  Respondent = c(1,2,3),
  Question1 = c("A","B","C"),
  Question2 = c("B","A","A"),
  Question3 = c("C","D","B")
)
counts <- as.data.frame(table(survey$Question1))
colnames(counts) <- c("Answer","Count")
ui <- fluidPage(
  titlePanel("Survey Results Dashboard"),
  fluidRow(
    column(6, plotlyOutput("barChart")),
    column(6, tableOutput("table"))
  )
)
```

```
server <- function(input, output){
  output$barChart <- renderPlotly({
    plot_ly(
      counts,
      x = ~Answer,
      y = ~Count,
      type = "bar"
    )
  })
  output$table <- renderTable({
    survey
  })
}
shinyApp(ui, server)
```



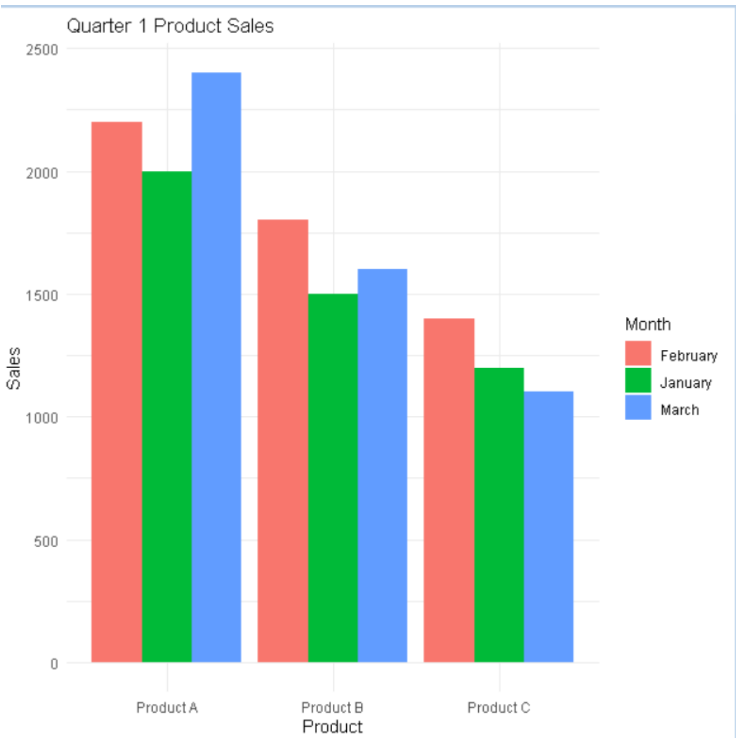
SET-15

Q1-Grouped Bar Chart

R PROGRAM:

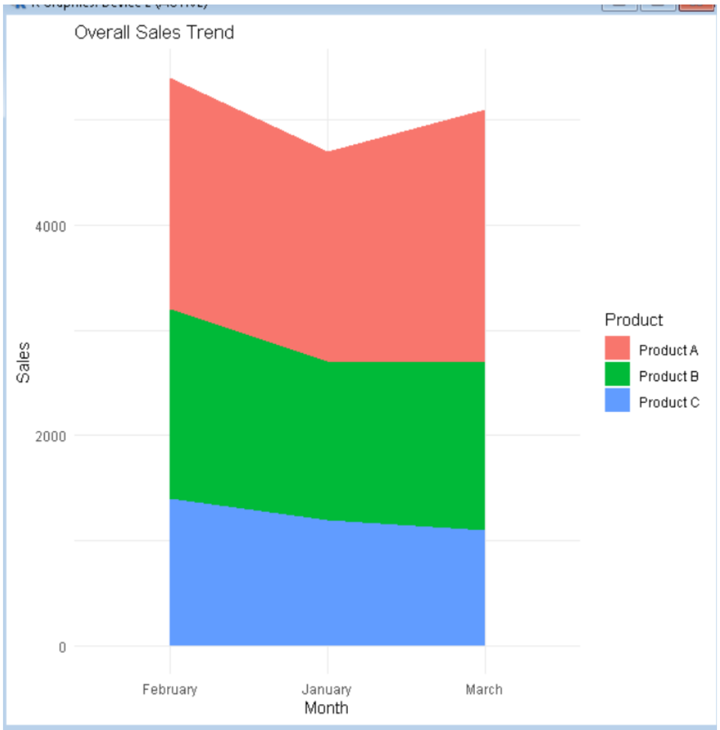
```
library(ggplot2)
library(tidyr)
sales <- data.frame(
  Product = c("Product A","Product B","Product C"),
  January = c(2000,1500,1200),
  February = c(2200,1800,1400),
```

```
  March = c(2400,1600,1100)
)
sales_long <- pivot_longer(
  sales,
  cols = January:March,
  names_to = "Month",
  values_to = "Sales"
)
ggplot(sales_long, aes(x = Product, y = Sales, fill = Month)) +
  geom_bar(stat = "identity", position = "dodge") +
  labs(
    title = "Quarter 1 Product Sales",
    x = "Product",
    y = "Sales"
  ) +
  theme_minimal()
```



Q2- Stacked Area Chart

```
library(ggplot2)
library(tidyr)
sales <- data.frame(
  Product = c("Product A","Product B","Product C"),
  January = c(2000,1500,1200),
  February = c(2200,1800,1400),
  March = c(2400,1600,1100)
)
sales_long <- pivot_longer(
  sales,
  cols = January:March,
  names_to = "Month",
  values_to = "Sales"
)
ggplot(sales_long,
       aes(x = Month,
           y = Sales,
           fill = Product,
           group = Product)) +
  geom_area() +
  labs(
    title = "Overall Sales Trend",
    x = "Month",
    y = "Sales"
  ) +
  theme_minimal()
```



Q3- Table

R PROGRAM:

```
sales <- data.frame(
  ProductID = c(1,2,3),
  ProductName = c("Product A","Product B","Product C"),
  JanuarySales = c(2000,1500,1200),
  FebruarySales = c(2200,1800,1400),
  MarchSales = c(2400,1600,1100)
)
print(sales)
```

```
+   MarchSales = c(2400,1600,1100)
+ )
>
> print(sales)
  ProductID ProductName JanuarySales FebruarySales MarchSales
1         1   Product A         2000          2200       2400
2         2   Product B         1500          1800       1600
3         3   Product C         1200          1400       1100
>
```

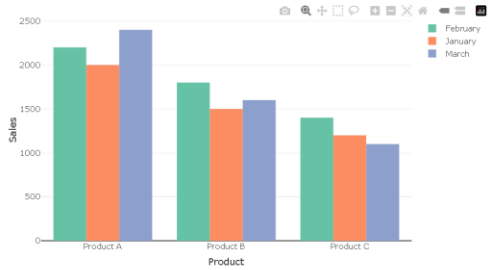
Q4- Dashboard (Shiny)

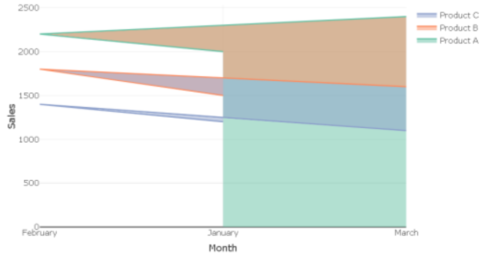
R PROGRAM:

```
library(shiny)
library(plotly)
library(tidyr)
sales <- data.frame(
  Product = c("Product A","Product B","Product C"),
  January = c(2000,1500,1200),
  February = c(2200,1800,1400),
  March = c(2400,1600,1100)
)
sales_long <- pivot_longer(
  sales,
  cols = January:March,
  names_to = "Month",
  values_to = "Sales"
)
ui <- fluidPage(
  titlePanel("Product Sales Dashboard"),

  fluidRow(
    column(6, plotlyOutput("barChart")),
    column(6, plotlyOutput("areaChart"))
  ),

  fluidRow(
    tableOutput("salesTable")
  )
)

server <- function(input, output){
```

```
  output$barChart <- renderPlotly({
    plot_ly(
      sales_long,
      x = ~Product,
      y = ~Sales,
      color = ~Month,
      type = "bar"
    )
  })
  output$areaChart <- renderPlotly({
    plot_ly(
      sales_long,
      x = ~Month,
      y = ~Sales,
      color = ~Product,
      type = "scatter",
      mode = "lines",
      fill = "tonexty"
    )
  })
  output$salesTable <- renderTable({
    sales
  })
}
shinyApp(ui, server)
```

Product Sales Dashboard





Product	January	February	March
Product A	2000.00	2200.00	2400.00
Product B	1500.00	1800.00	1600.00
Product C	1200.00	1400.00	1100.00